

Terahertz Time-Domain Spectroscopy of $0.73\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{--}0.27\text{PbTiO}_3$ Single Crystal

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Lead magnesium niobate titanate is an important ferroelectric material. In this study, the terahertz (THz) transmission properties of a $0.73\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{--}0.27\text{PbTiO}_3$ single crystal were investigated using a time-domain spectroscopy method. Complex refractive index and dielectric dispersion functions were determined from the amplitude and phase information derived from time-domain responses. Based on calculations, it was concluded that the room-temperature dielectric constant of the single crystal equal to ~ 30 at 1 THz. This result could be a useful reference for development of ferroelectric-material-based THz components and devices.

I. Introduction

SOLID solutions of lead magnesium niobate–lead titanate (abbreviated as PMN–PT) have attracted extensive interest mainly due to their excellent dielectric, ferroelectric, piezoelectric and pyroelectric properties, and potential applications in the electronic industry.^{1–11} At the morphotropic phase boundary (with PMN: PT (70:30), for example, a PMN–PT single crystal presents an extraordinarily large electromechanical response (piezoelectric coefficient of ~ 2500 pC/N) and a low hysteresis behavior, which makes it distinct from and superior to other piezoelectric materials in making high-performance sensors and actuators.^{2,12–14} PMN–PT is also attractive for its puzzling microstructure and rich phase transitions, which lead to complicated dielectric behaviors. In the literature, experiments have been conducted to determine the dielectric constant of PMN–PT over a frequency range from tens of hertz to that of visible light.^{1,6,8,15–17} At low frequencies (up to MHz), PMN–PT (both single crystals and polycrystalline) behave either like a normal ferroelectric crystal (when the content of PT is large) or like a relaxor (when the content of PT is small) mainly due to complex domain structures, with a typical dielectric constant of several thousand at room temperature.^{10–13} As the frequency increases, the dielectric constant of PMN–PT gradually decreases, followed by a fast drop when the frequency approaches the GHz range, due to the reduced contribution from ferroelec-

tric domains and other structure features. In the infrared and visible frequencies (i.e. the electro-optical behaviors) the value of the relative dielectric constant becomes even smaller.^{18–20} The dielectric constant as a function of the direct current (DC) bias, temperature, and time has also been extensively studied.^{21–23} Nevertheless, there are few experimental attempts on the dielectric measurement of PMN–PT in the terahertz (THz) range.

THz radiation is an electromagnetic wave with frequency falling in between microwave and infrared radiations. THz technology is believed to hold promises in medical imaging, security detection, nondestructive testing, etc.^{24,25} Due to technical difficulties in generation, detection and modulation of THz waves, however, the study of dielectric behaviors of ferroelectric materials in this frequency range remains rather limited, despite of preliminary results of a few ferroelectric materials (including barium titanate and strontium titanate).^{26–31} More extensive results are required to bridge the gap between microwave and infrared data and to study the soft mode behaviors of these materials. In light of the situation, we chose to work on the THz dielectric characterization of a single crystal of $0.73\text{Pb}(\text{Mg}_{1/3}\text{Nb}_{2/3})\text{O}_3\text{--}0.27\text{PbTiO}_3$ using the THz-TDS method.

II. Experiments

Carefully polished PMN–27PT single crystals of sizes $5\text{ mm} \times 5\text{ mm} \times 0.4\text{ mm}$ and $5\text{ mm} \times 5\text{ mm} \times 0.2\text{ mm}$ were used in the experiment. THz transmission measurements were performed at room temperature by using a THz time-domain spectroscopy system (from Ekspla in Lithuania), pumped by a Ti: Sapphire femtosecond laser (Coherent, CA), the working principle of which has been described in the literature.²⁶ The THz emitter and detector features a microstrip photoconductive antenna fabricated on a low-temperature grown GaAs substrate.

III. Results and Discussion

The transfer function of a single plate by a normal incidence of THz ray can be formulated as³²:

$$T(\omega) = \frac{4N}{(1+N)^2} \times e^{-i(N-1)cd/c} \times \text{FP}(\omega) \quad (1)$$

where N and d are the complex refractive index and the thickness of the sample plate, ω and c are the angular frequency and the speed of the light, $\text{FP}(\omega)$ is the contribution

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of Fabry–Perot effects for low absorptive or thin samples, respectively. The transmittance T can be further resolved into a transmission amplitude A and a phase delay ϕ , following $T = Ae^{i\phi}$. Also the complex refractive index N can be expressed in terms of the refractive index n and the extinction coefficient κ , as $N = n - i\kappa$. The relationship between these values can be derived from Eq. (1) as follows:

$$\phi = \tan^{-1} \left(\frac{\kappa(n^2 + \kappa^2 - 1)}{n[(n+1)^2 + \kappa^2] + 2\kappa^2} \right) - \frac{(n-1)\omega d}{c} + 2p\pi, \quad (2)$$

where p is an arbitrary integer, and

$$\ln A = \ln \left[\frac{4\sqrt{n^2 + \kappa^2}}{(n+1)^2 + \kappa^2} \right] - \frac{\kappa\omega d}{c}. \quad (3)$$

For nondispersive materials, Eqs. (2) and (3) are both linear with respect to ω , with slopes proportional to $(n-1)$ and κ , respectively. These equations are accurate provided: (1) the generated THz light is a plane wave; (2) both surfaces of the sample plates are optically smooth; (3) the air gaps adjacent to the two ends of samples are long enough and nonabsorptive; and (4) the samples are highly absorptive or thick so that the Fabry–Perot effect can be ignored.

The results acquired from THz-TDS are shown in Fig. 1, for PMN–PT and air reference, respectively. In Fig. 1(a), the signal to noise ratio is relatively low because of the high absorption of the sample. The original pulse is modulated into two pulses, where the first one is the main pulse, and

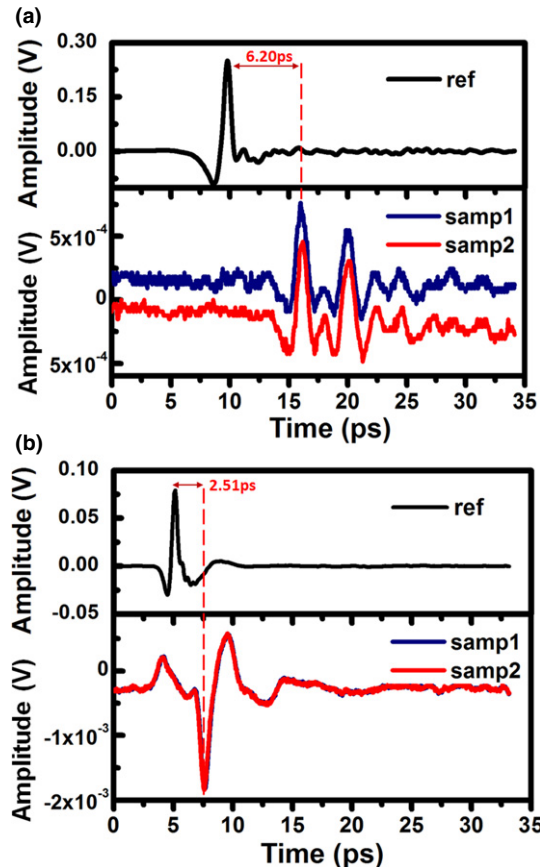


Fig. 1. The time-domain responses of reference (air) and samples (0.73PMN–0.27PT single crystals) with thickness of (a) 0.4 mm, (b) 0.2 mm, respectively. Note that there are two curves in the lower graphs of both (a) and (b), and in (b) the two curves almost overlap.

the second one may be subjected to a weaker birefringence effect^{33,34} (Fabry–Perot effects can be excluded because of the high absorption of the sample plates). A slight DC shift between the two curves can be ignored because it is irrelevant to spectral analysis. In contrast to the thicker sample, measurements on the thinner sample in Fig. 1(b) show lower noise, and perfect consistency.

One method for estimating the refractive index n can be derived by considering the time delay Δt of the peak signal caused by the higher refractive index of the sample. The relationship is written as $n = c\Delta t/d + 1$, where the thickness d is 0.4 and 0.2 mm for the two sets of samples. From Fig. 1(a), the time delay (for the main pulse) is 6.20 ps, whereas from Fig. 1(b) it is 2.51 ps, resulting in a value of 5.65 and 4.77, respectively.

The frequency responses are derived from time-domain data by using fast Fourier transform algorithm. As shown in the Fig. 2, the amplitude information is shown in the log scale, whereas the phase data are shifted by multiples of 2π to ensure that the absolute differences between adjacent values are no more than π . The phase data show an obvious linear dependence at a wide frequency range, which strongly correlates with the second term in Eq. (2). Poor linearity out of the frequency range is possibly due to the high dispersion of the material. The linear parts of the data are fitted by dashed lines, the slopes of which are compared with Eq. (2),

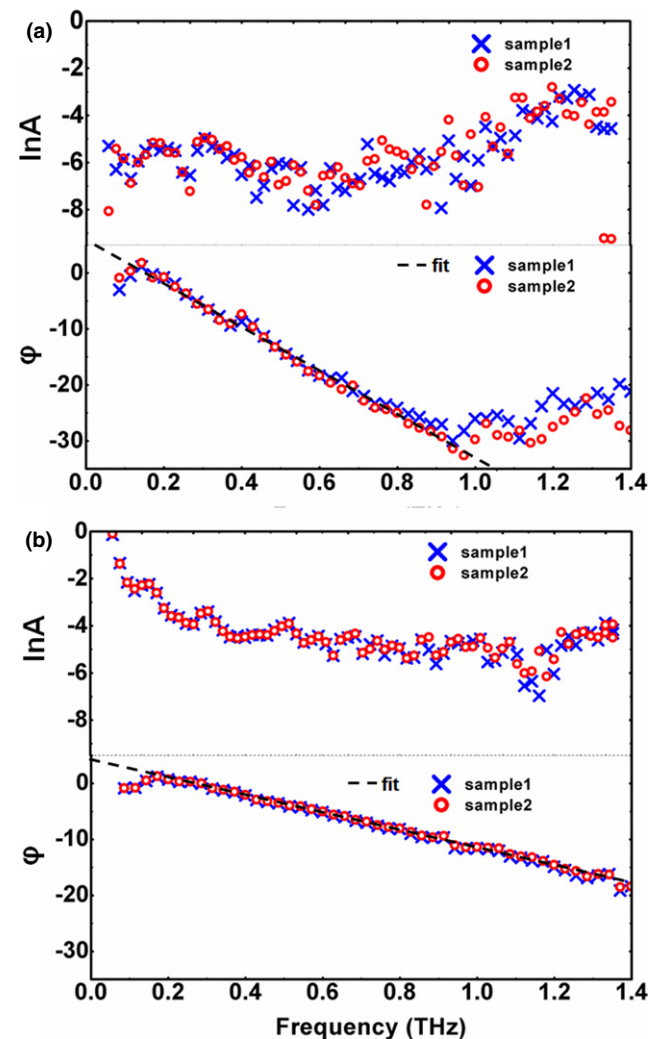


Fig. 2. The frequency-domain responses of two sets of samples (0.73PMN–0.27PT single crystals) with thickness of (a) 0.4 mm, and (b) 0.2 mm, respectively. The lower graphs of both (a) and (b) are shifted by a factor of 2π so that they show a linear feature. The dashed lines are linear fits of the phase data.

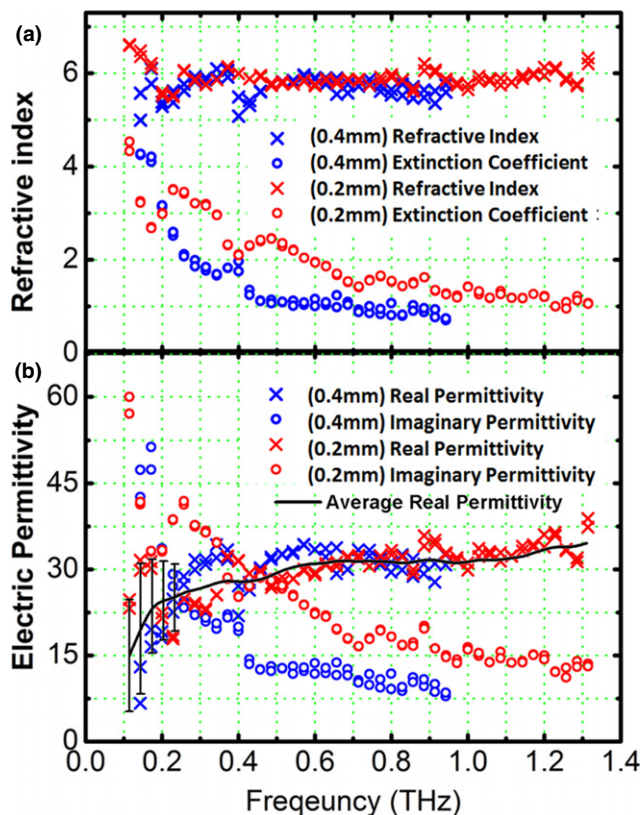


Fig. 3. The retrieved (a) complex refractive index spectra and (b) dielectric function of 0.73PMN-0.27PT single crystal. Blue (red) marks are for samples of 0.4 mm (0.2 mm) thickness, respectively. The average real permittivity is shown in (b) by a solid line, with error bars representing the standard deviations (only for frequencies <0.23 THz).

as $s = 2\pi(n-1)d/c$. We found $s = 39.11$ rad/THz from Fig. 2(a), and $s = 15.62$ rad/THz from Fig. 2(b), resulting in another pair of estimation of n , 5.67 and 4.73. These results are consistent with previous ones derived from time delay.

The refractive indices are further substituted into Eq. (2), together with values of log amplitudes to estimate the extinction coefficient κ , which is solved using an inverse problem solver “trust-region-dogleg” algorithm built in Matlab software. After that, the two sets of estimations of n and κ are applied as initial values in solving another inverse problem governed by Eqs. (2) and (3). To define the convergence condition, we employ an error function δ for simultaneously fitting the log amplitude and the phase delay as $\delta(n, \kappa) = (\phi - \phi')^2 + (\ln A - \ln A')^2$, where ϕ and $\ln A$ are calculated from Eqs. (2) and (3) by substituting the initial values of n and κ , whereas ϕ' and $\ln A'$ are the measurement values, as those presented in Fig. 2.³² This time we apply an Levenberg-Marquardt algorithm, because the target equation system is nonlinear.

The calculated results of n and κ are shown in Fig. 3(a). Across the frequency range from 0.23 to 1.3 THz, the retrieved values of n for the four samples are lying at 5.0 ± 1.0 . The values of κ above 0.7 THz are mainly at 1.0 ± 0.3 , and are slightly higher at lower frequencies. The four sets of data are roughly matched with each other at ranges 0.23 ~ 1.3 THz, with small deviations. The deviations are attributed to various disturbances in the measurements, such as scattering effects from the roughness of sample surfaces, or deviations from perfect plane wave condition. The complex dielectric constant $\epsilon = N^2 = \epsilon' - i\epsilon''$ can also be derived, whose real and imaginative parts are shown in Fig. 3(b). Due to the large divergence, the lower and higher ends of the dielectric function are not included in the calculations. It was found that the value of ϵ' of our sample is ~ 30 .

For perovskite ferroelectrics, the complex dielectric function is governed by phonon vibrations in infrared range, and polar nanoregion (PNR) relaxations in microwave range. Our inspection range (0.1 ~ 1.4 THz) is at the intersection of these two ranges. Previous studies of broadband spectroscopy²⁰ suggested that, for frequencies near 1 THz, the TO₁ phonon mode (soft mode) is most critical. This mode describes mainly the vibration of Pb cations against the BO₆ octahedra framework, and is generally split into two modes of A₁(TO₁) (~2.3 THz) and E(TO₁) (~0.7 THz) due to the rhombohedral symmetry of PNRs at room temperature. At lower ends (~30 GHz), a relaxation mode R₁ due to the expansion/shrinking of PNRs also has significant effects on the THz range responses. Due to the complicated mode characteristics, the dielectric constant of PMN-PT can be very sensitive to the composition, crystallinity, and temperature.²⁰ These may explain why ϵ of our PMN-27PT is in same order of but not exactly equal to the ϵ (~50) for PMN-29PT in the literature.¹⁸ One possible way to improve the measurement accuracy could be through using ultrathin sample. For example, BaTiO₃ single crystal with a thickness of 32 μ m (prepared by H₃PO₄ etching) was used in the literature.²⁶ For PMN-PT single crystal, however, it is difficult to thin it to this level because of its poor mechanical strength. It is also important all the measurement methods involve multistep data treatments which inevitably could introduce some error.^{26-29,35} How to further improve the THz measurement accuracy is still in progress.

IV. Conclusion

In summary, we have performed THz-TDS study on a PMN-0.27PT single crystal. The dielectric constant of the material was found to be ~ 30 at 1 THz. This result as well as the measurement methodology could be used as a reference for the future study of THz dielectric behaviors of ferroelectric materials.

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